LA-8: Practice Data Analysis (15 points)

Learning Outcomes

In this assignment, you will practice the data analysis skills that you have learned this semester:

- Recoding variables with `mutate()` and `case_when()` (LA-5 to LA-7).
- Calculating descriptive statistics for groups with `descr()` (LA-4 to LA-7).
- Selecting, conducting, and interpreting a statistical test (LA-7).
- Making a graph with `ggplot2` (LA-6).

> **Tip**
>
> Read all the instructions carefully before starting the assignment. Set up your R script as you did in LA-4 to LA-7 (name, uNID, date, assignment number).

> **Important**
>
> **How to report your answers.** Write each answer as a comment in your R script, labeled with its question number (e.g., `# 3a: ...`). Report test results in APA format, as shown in the section "Conducting a Statistical Test in R" at the end of this document. Round means and standard deviations to two decimal places.

> **Note**
>
> **Key ideas (from LA-7):**
>
> - The **independent variable (IV)** is the variable we think *explains* or *predicts* differences. The **dependent variable (DV)** is the outcome we are measuring.
> - A **categorical** variable sorts cases into groups. A variable coded with numbers like 0 and 1 can still be categorical if the numbers stand for groups (like `DonorStatus` in LA-6). A **continuous** variable is a number that can take many values.
> - Use the flowchart in the section "Selecting a Statistical Test" at the end of this document to choose a test. Start at the top with the DV.
> - In this course, if $p < .05$, we conclude that there **is** a statistically significant difference (or relationship). If $p \geq .05$, we conclude that we **do not** have evidence of a difference (or relationship).

Instructions

1) Load the packages below. Then download the full COVID dataset (`covid-full.csv`) and its codebook (`covid-full-codebook.csv`) from Canvas, upload the dataset to your Posit Cloud project (as you did in LA-4 to LA-7), and read it into R. Use the codebook to familiarize yourself with the variables in the data.

- `tidyverse`
- `summarytools`
- `rstatix` (if R says it can't find the package, run `install.packages("rstatix")` first)

> **Note**
>
> **About this dataset.** This is a larger version of the dataset you used in LA-7 (291 rows and 29 countries), so your numbers will not match LA-7. As before, each row (case) is **one country on one survey date**, and the values are **averages** of the people surveyed in that country on that date. For example, `handwashes` is the average number of times people in that country washed their hands on the previous day.

2) Create a new variable called `asia_europe`. Have the variable equal `Asia` if the country is China, India, Japan, Singapore, or South Korea. Have the variable equal `Europe` if the country is Denmark, France, Germany, Italy, or Spain. Run a frequency distribution of `asia_europe` to check it.

   Then, obtain descriptive statistics for the number of times respondents washed their hands on the previous day (`handwashes`) for Asia and Europe.

   a) What is the mean and standard deviation of handwashes in Asia?
   b) What is the mean and standard deviation of handwashes in Europe?

> **Tip**
>
> Use `mutate()`, `case_when()`, and `|` ("or") the same way you created `region` in LA-7. Remember to save the result back to `covid` (start with `covid <- covid |> ...`), because you will use `asia_europe` in Questions 3, 5, and 7.
> Only 10 of the 29 countries are in Asia or Europe as defined above. All other countries (e.g., United Kingdom, Taiwan) will be `NA` in `asia_europe`. That is expected, and R leaves them out of the statistical tests.
> To get descriptive statistics for each region, use `group_by()` before `descr()`, as in LA-7, or use `filter()` and `descr()` for each region.

3) Conduct a statistical test to determine whether there is a significant difference in the average number of times people in Asia compared to Europe washed their hands.

   a) State your hypothesis.
   b) Select and conduct a statistical test to address your hypothesis.
   c) Report the test statistic and p-value in APA format.
   d) Does the statistical test support your hypothesis? Explain.

> **Tip**
>
> As in LA-7, `t_test()` subtracts the mean of the second group from the mean of the first group (groups are in alphabetical order). A **negative** t means the first group's mean is **lower**. Use your answers to Question 2 to see which region is higher.

4) Select and conduct the appropriate statistical test to determine whether there is a significant linear relationship between how many times people left their home (`times_left_home`) and the total number of contacts they had with other people (`total_contacts`).

   a) State your hypothesis.
   b) Select and conduct a statistical test to address your hypothesis.
   c) Report the test statistic and p-value in APA format.
   d) Does the statistical test support your hypothesis? If there is a significant relationship, describe it (e.g., is it positive or negative?). Does this finding make sense? Why or why not?

> **Tip**
>
> `times_left_home` has fewer distinct values than `total_contacts`, but it is an average (e.g., 1.4 times), so it is continuous. When you are looking for a relationship between two continuous variables, it does not matter which variable you list first in `cor_test()`.

5) Is a country being in Asia or Europe related to whether most of its respondents said their lives were greatly impacted by COVID (`majority_life_impacted`)?

   a) State your hypothesis.
   b) Select and conduct a statistical test to address your hypothesis.
   c) Report the test statistic and p-value in APA format.
   d) Does the statistical test support your hypothesis? If there is a significant difference, describe it (e.g., which region, Asia or Europe, had more survey dates on which most respondents said their lives were greatly impacted by COVID?).

> **Tip**
>
> `majority_life_impacted` is coded 0 and 1, but the numbers stand for groups (1 = most respondents said their lives were greatly impacted; 0 = they did not).
> The chi-squared command uses `covid$asia_europe`, which means "the `asia_europe` column of the `covid` dataframe":
> ```
> chisq_test(x = covid$asia_europe, y = covid$majority_life_impacted)
> ```
> For 5d, use `filter()` and `freq()` to compare the regions, e.g.:
> ```
> covid |>
>   filter(asia_europe == "Asia") |>
>   freq(majority_life_impacted)
> ```
> Then repeat for `Europe`.

6) **BONUS:** Use the data and a statistical test to determine whether there is a relationship between the number of days since the COVID outbreak began (`days_since_outbreak`) and how often respondents wore masks outside (`masks_outside`, a score from 0 to 100). Conduct the statistical test, report the results of the test (i.e., test statistic and p-value) and explain your conclusion.

7) **BONUS:** Create a bar chart showing the average number of times people living in Asia and Europe washed their hands. Remove the x-axis label so that only the tick marks are labeled (i.e., with "Asia" and "Europe") and rename the y-axis label to "Average handwashes (previous day)".

> **Tip**
>
> `geom_col()` **adds up** all the values in each group, so it would show totals, not averages. To plot the average, use `stat_summary(fun = mean, geom = "col")` instead. Use `filter(!is.na(asia_europe))` to remove countries that are not in Asia or Europe, and `labs()` to change the axis labels:
> ```
> covid |>
>   filter(!is.na(asia_europe)) |>
>   ggplot(aes(x = asia_europe, y = handwashes)) +
>   stat_summary(fun = mean, geom = "col") +
>   labs(x = ..., y = ...)
> ```

Submission

Submit your R script (named `LA-#_FirstName-LastName.R`) to Canvas.

Your R script should:

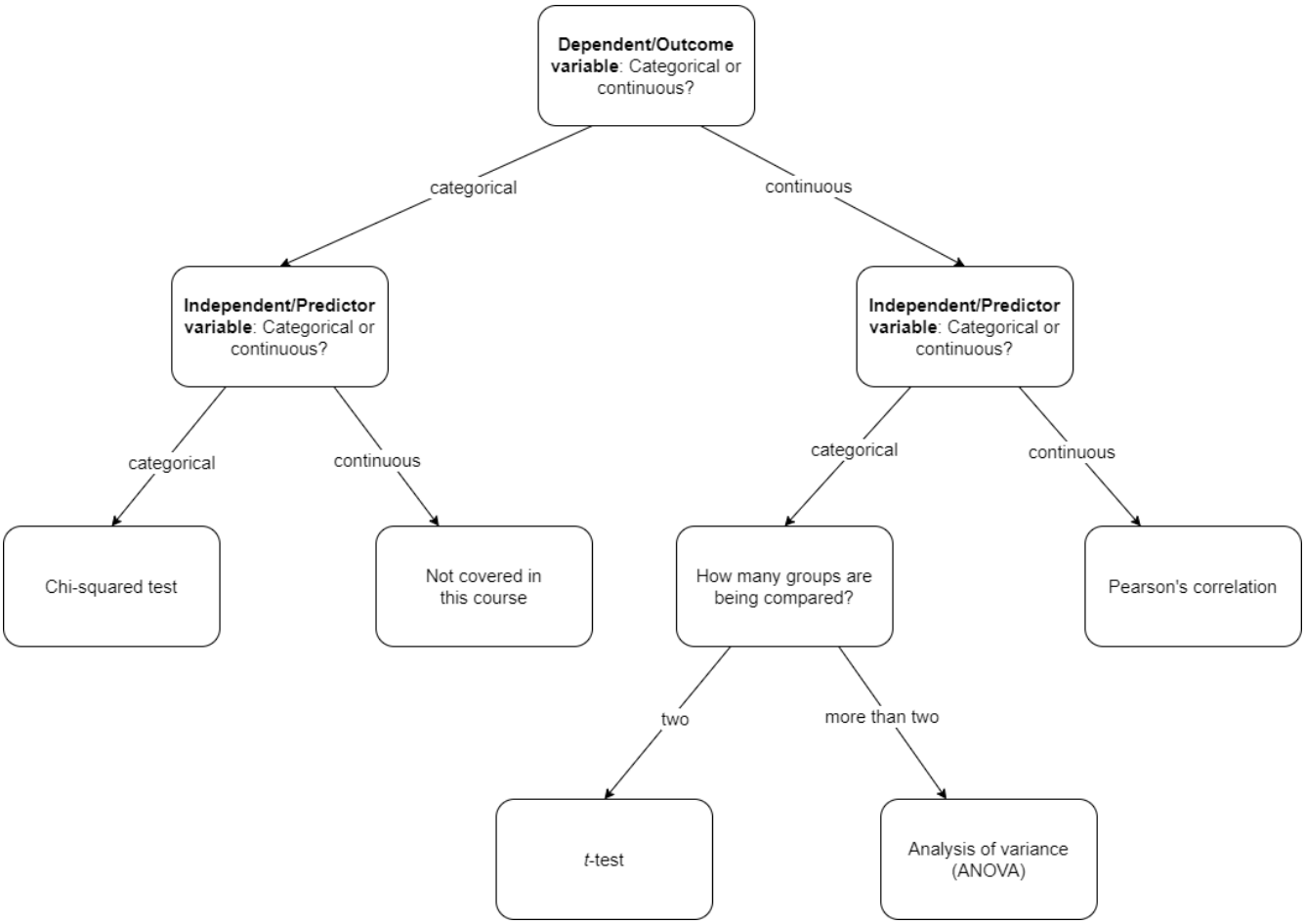
1) Include commands and functions that are necessary to address all the questions in the assignment.
2) Contain comments that answer the questions in the assignment.
3) Run in its entirety without errors.

To ensure that your R script runs without errors, you should:

- Save your script.
- Navigate back to Your Workspace on Posit Cloud.
- Reopen your project.
- Run the entire script line-by-line without editing it to ensure there are no errors.

> **!** Important
>
> These standards apply to all submissions in this course that require R scripts. You should follow these instructions for preparation, naming, and saving of your R script for all of your individual lab assignments.

Selecting a Statistical Test



Conducting a Statistical Test in R

The example output below is **made up** for illustration, so your numbers will be different. Use these examples to see what the output looks like and how to report it.

Chi-squared test

A chi-squared test is used when we want to compare two categorical variables. To do so in R, you can use the following command (`df`, `dep_var`, and `ind_var` have to be replaced with the names of your dataframe, dependent variable, and independent variable, respectively). The output that results from running this command in R is shown below as comments.

```
chisq_test(x = df$ind_var, y = df$dep_var)

# A tibble: 1 × 6
#       n statistic         p    df method          p.signif
# * <int>     <dbl>     <dbl> <int> <chr>           <chr>
# 1   200      6.84    0.0327     2 Chi-square test *
```

To report the results of your chi-squared test in APA format, you would use the following format, replacing the

text in bold font with the results from your Console:

$\chi^2(\mathbf{df}, N = \mathbf{n}) = \mathbf{statistic}, p = \mathbf{p}$

From the results of the above chi-squared test, I would report results like this: $\chi^2(df = 2, N = 200) = 6.84, p = .033$. If the p-value is less than .001, you do not need to report the exact value. If it is greater than .001, then report the exact value rounded to two or three decimal places.

t-test

A t-test is used to compare the mean values of a continuous variable between two groups. To do so in R, you can use the following command (`df`, `dep_var`, and `ind_var` have to be replaced with the names of your dataframe, dependent variable, and independent variable, respectively). The output that results from running this command in R is shown below as comments.

```
df |>
        t_test(dep_var ~ ind_var)

# A tibble: 1 × 10
#   .y.        group1 group2    n1    n2 statistic    df        p    p.adj p.adj.signif
# * <chr>      <chr>  <chr>  <int> <int>     <dbl> <dbl>    <dbl>    <dbl> <chr>
# 1 dep_var ind_var1  ind_var2  48    52     -4.12  97.6 7.94e-05 7.94e-05 ****
```

To report the results of your t-test in APA format, you would use the following format, replacing the text in bold font with the results from your Console:

$t(\mathbf{df}) = \mathbf{statistic}, p = \mathbf{p}$

Using the results above, I would report the results of a t-test like this: $t(df = 98) = -4.12, p < .001$. Round `df` to a whole number if it has decimals.

Analysis of variance (ANOVA)

ANOVA is used when we want to compare the mean values of continuous variable between more than two groups. To do so in R, you can use the following command (`df`, `dep_var`, and `ind_var` have to be replaced with the names of your dataframe, dependent variable, and independent variable, respectively). The output that results from running this command in R is shown below as comments.

```
df |>
        anova_test(dep_var ~ ind_var)

# ANOVA Table (type II tests)
#
#   Effect  DFn DFd     F        p p<.05   ges
# 1 ind_var   2 450 8.936 0.000156     * 0.038
```

To report the results in APA format:

F (`DFn`, `DFd`) = `F-value`, p = `p-value`

Using the results above, I would report: $F(2, 450) = 8.94, p < .001$.

Correlation

We use Pearson's correlation to examine a linear relationship between two continuous variables. To run a correlation in R, you can use the following command (`df`, `dep_var`, and `ind_var` have to be replaced with the names of your

dataframe, dependent variable, and independent variable, respectively). The output that results from running this command in R is shown below as comments.

```
df |>
        cor_test(dep_var, ind_var)

# # A tibble: 1 × 8
#   var1          var2                cor statistic        p conf.low conf.high method
#   <chr>         <chr>             <dbl>     <dbl>    <dbl>    <dbl>     <dbl> <chr>
# 1 dep_var       ind_var            0.31      4.59 0.00000793   0.178     0.431 Pearson
```

To report the results in APA format:

> Pearson's $r =$ `cor`, $p =$ `p-value` (or $p < .001$ if the p-value from your test is very small, i.e., less than .001)

Using the results above, I would report: Pearson's $r = 0.31$, $p < .001$.